

Heavy Metals and Toxic Effect of Fish Labia Rohita

College students find it difficult to understand the subject taught mainly due to understanding standard references. While preparing this THE TEXTBOOK OF HEAVY METALS AND TOXIC EFFECT OF FISH Labia rohita (Analysis of Textile waste in freshwater bodies of Native Place in Namakkal District) has been prepared to serve the special purpose of college students. This book explains the advanced Research in toxicology and the effect of heavy metals on fish in a simplified manner. This book can be used for UG and PG zoology students, who are looking for a quick revision before the semester Examination. This book has been written for students and preparation in the field of fishery biology, no such book covers a suitable original picture with needed explanation, General principles of toxicology studies and creates awareness to public and practice.



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Kalaimani Subramani

Heavy Metals and Toxic Effect of Fish Labia Rohita

Analysis of Textile waste in fresh water bodies of
Native Place in Namakkal District

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**ANALYSIS OF TEXTILE WASTE IN FRESH WATER BODIES
OF NATIVE PLACE IN NAMAKKAL DISTRICT AND TOXIC
EFFECT OF FRESH WATER FISH *Labia rohita***

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CHAPTER - I

INTRODUCTION

According to Mason (1991), there are five major types of toxic pollutants known to man: Metals arising from industrial process and some agricultural applications (lead, copper, nickel and zinc). Organic compounds, originating from industrial, agricultural and some domestic sources (herbicides, PCB's, organochloride pesticides, chlorinated aliphatic hydrocarbons, organo metallic compounds and phenols), Gases (ammonia and chlorine) Anions (cyanides, fluorides, sulphides and sulphites) v), acids and alkalis. Amongst the pollutants contaminating water bodies, metals play an important role (Witeska et al., 1995). As mentioned in chapter one, answers are always sought, in practice, to the problems raised by water pollutions, especially when it comes to metal pollution (Nussey, 1998)

Metals are elements found naturally in aquatic ecosystems due to various processes such as weathering and erosion (Viljoen, 1999). Some of these metals are essential to living organisms in trace amounts (for example copper and zinc). Essential trace elements have a narrow optimal concentration range for growth and reproduction, and both excess and shortage can be detrimental to organisms (Pelgrom et al., 1994), with unusually high concentration becoming toxic to aquatic organisms (Wepener, et al., 2001). Metals are present in very low concentrations in natural aquatic ecosystems (Nussey, 1998), usually at the nanogram to microgram per litre level, but recently the occurrence of especially heavy metals in excess of natural loads, has become an increasing concern (Biney et al., 1994; Bennet-Chambers et al., 1999) for aquatic ecosystem. Heavy metals are part of a group of elements, whose hydro chemical cycles have been accelerated to a great extent by man (Viljoen, 1999). The most important heavy metals in water pollution are zinc, copper, lead, cadmium, mercury, nickel and chromium (Abel, 1989; Seymore, 1994; Viljoen, 1999). Metal uptake by aquatic organisms is a

two-phased process, firstly involving rapid adsorption or surface binding, followed by a slower transport into the cell Interior. Transport of metals into the intracellular section may be aided by either diffusion of the metal ion across the cell membrane or by active transport by a carrier protein (Brezonik et al., 1991; Wepener et al., 2001).

Increase In Heavy Metals In Natural Ecosystem, Metals are introduced into the environment by a wide range of natural and anthropogenic sources (Wepener, et al.,2001) and with anthropogenic being either domestic or industrial (Biney et al., 1994). Heavy metals are often present at elevated concentrations in aquatic ecosystems, due to the rapid growth in population (Biney et al., 1994: Seymore, 1994), an increase in industrialization (Biney et al., 1994: Pelgrom et al., 1994), the increase of urbanization and socio-economic activities, exploration and exploitation of natural resources, 4) Extension of irrigation and other modern agricultural practices, as well as 5) The lack of environmental regulations (Biney et al., 1994). Consequently aquatic organisms are exposed to the elevated levels of metals (Pelgrom et al., 1995a) levels not previously encountered (Nussey, 1998), posing a great threat to aquatic organisms in particular, and to the whole ecosystem in general (Zou and Bu, 1994; Zou, 1997). An accelerated release of heavy metals into the aquatic environment poses serious water pollution problem because of their toxicity (Tulasi et al., 1989), persistence and bioaccumulation in food chains (Rishi and Jain, 1998). Due to their adverse effects on aquatic ecosystems, it is important to identify the sources and measure the discharge of heavy metals into systems nationwide (Biney et al., 1994). Metals and pesticides, in particular, have an inclination to accumulate and undergo food chain magnification (James et al., 1998). It is thus also important to monitor the bioaccumulation of these metals in a system, in order to assess the possible impact on human health (fish consumed), and organism health (exposure to a pollutant or consumption by predators) (Kotze et al., 1999). Metals pollution is an important worldwide

problem, which is growing at an alarming rate.

Toxicity of heavy metals:

Unlike other classes of pollutants, which can be biodegraded and destroyed completely, metals are non-biodegradable (Wepener, et al., 2001) and can neither be created nor destroyed. The Internal organization of a metal can play in its transfer to higher trophic levels, by affecting the assimilation effectiveness by a consumer (Munger et al., 1999). Even though metal elimination routes are more abundant than uptake routes, metal accumulation occurs more rapidly than removal, probably due to the presence of metal binding proteins in the tissues of aquatic organisms (Kargin and Cogun, 1999).

Monitoring of heavy metal effects on aquatic environments and living organisms:

The random discharge of various waste, pollute most ecosystems and affect survival and physiological activities of organisms in these systems (James et al., 1998). Free metal ions cause more serious damaging effects on aquatic organisms than their more complex (James et al., 1998). The duration of exposure of a specific concentration of toxin can influence whether it will kill an aquatic organism (Skidmore, 1964). The health of fish may be affected, either directly through uptake from the water. Or indirectly through their diet of vegetarian, invertebrates or smaller fish (Kime et al., 1996). Metals released into aquatic ecosystems, are responsible for several fish physiology irregularities (Sehal and Saxena, 1986). They can also disturb the ion regulatory mechanisms in aquatic organisms (Hansen et al., 1996). Aluminums have been classified as being one of the most toxic metals in the aquatic environment. McCormick and Jensen (1989) chose to use aluminum as a co- factor with low pH in exposures to fathead minnow, because it is frequently elevated acidified aquatic environment. Chromium: Stouthart et al., (1995) tested the effect of water pH on chromium toxicity to early life stages of the *Labio rohita* eggs, after fertilization, were immediately exposed to Cr (VI) concentrations

of 3.9 and 9.6 m mol/l at water pH 7.8 and 6.3. No effects of chromium were observed at pH 7.8. The hatching success of the carp eggs wasn't affected at all, due to the chorion and perivitelline fluid being an effective barrier for the Cr (VI) during the egg stage. Lead: Ozoh (1979b) found that intraperitoneal lead toxicity affected ovulation and its frequency in adult *Labio rohita*. Egg mortality was high influencing hatching success. Lead induced abnormalities in fry included; deformities in the brain, caused lack of eye formation (anophtalmia) and some cases of reduced eyes (microphthmia), malformations of the head regions, and blood circulation problem for example no blood circulation even if the heart was beating. Poor yolk Resorption resulted in yolk cells being seen even when covered by integuments. Mercury: Weis and Weis (1991) observed a wide variety of eye and head defects in the mummichog, *Fondles heteroclitus*, were found after exposure to organic mercury. Cyclopia and even lack of eye formation or anophtalmia were just some of the abnormalities present. A reduction in the harshness of these defects was noted, as embryos were exposed at later developmental stages. It is a possibility that thus might have been a reflection on the time that it took for mercury to penetrate the chorine, as older embryos were not as vulnerable as the younger ones. Nickel: Nickel toxicity to aquatic life depends on the species, pH, water hardness and other environmental factors (Blaylock and Frank, 1979). According to Nebeker et al., (1985), nickel has been shown as moderately toxic to fish and aquatic invertebrates when compared to other metals. Newly fertilized rainbow trout eggs, exposed to 0.035 mg/l nickel, and produced smaller larvae than when eyed eggs or pre-swimup larvae were exposed even up to 0.134 mg /Zinc Biological importance of zinc: Zinc is a biologically essential nutritional trace element in minute quantities, which is ever-present in plants, microorganisms and animals (Hilmy et al., 1987). Zinc is known to be toxic to aquatic organisms in larger amounts (Skidmore, 1964; Holcombe et al., 1979; Crespo and Balasch, 1980). The mechanisms for zinc uptake across

membrane are through passive transport or by Endocytosis (Viarengo, 1985), or via the food (Heath, 1987). Effects of zinc on fish: The few available reports on the effects of zinc on the growth of fish are conflicting. Chapman (1978) determined that the growth of sockeye salmon (*Oncorhynchus nerka*) was slightly stimulated by zinc concentrations as high as 242 mg/liter. On the other hand a decrease in growth rate was observed in flag fish (Spehar, 1976), fathead minnows (Brungs, 1969), guppies (*Lebistes reticulatus*) (Crandall and Goodnight, 1962) and the minnow *Phoxinus phoxinus* (Bengtsson, 1947a). Copper: Biological importance of copper: Copper is an essential trace element to plants, animals and even humans (Dwaf, 1996). It is in fact a required element for all living organisms, and although the absolute concentration of copper is usually low in nature, it occurs in adequate quantities for growth in all aquatic environments (Stemann Nielsen et al., 1976). Cadmium: Biological importance of cadmium: Cadmium is a non-essential element, with no known biological function (Viarengo, 1985; Allen, 1994; Martinez et al., 1999), naturally found at low concentrations in natural waters (Eaton, 1974; Allen, 1994; Bennet-Chambers et al., 1999). It can have severe toxic effects on aquatic organisms when present in excessive amounts (Hollis et al., 1999), or even if present in extremely low concentrations (Witeska et al., 1995). Effects of cadmium on fish are generally very sensitive to cadmium poisoning, but certain species, for example roach, perch and tilapia, can tolerate high concentration of cadmium (Pelgrom et al., 1997). Calcium uptake suppression is one of the key mechanisms of cadmium to fish (Rattner and Heath, 1995). Cadmium can disrupt ionic balance by altering the permeability of the (Lee et al., 1996). Dietary cadmium uptake causes ionic disturbances of the blood influencing the kidney (Pratrap and Wendelaar Bonga, 1993).

AIM & OBJECTIVES

1. To analysis of textile waste effluent of fresh water bodies in native places in Namakkal District.
2. To study toxic effect of heavy metals on fresh water fish *Labia rohita* in selected areas.
3. To create awareness the effect of heavy metals to the local fisher man.

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CHAPTER II

MATERIALS AND METHODS

To catch adult *Labia rohita* from polluted Sunnambu odai pond (21.1 cm and 138.4 g). Ten fish from each station were used for analysis. The sizes and weights of the animals did not vary significantly from station to station ($P>0.07$). Station Sunnambu odai is a relatively clean area of the river system because it is north of poultry industrial and agricultural areas and is therefore relatively less affected from contaminated poultry farms effluents. The other stations, however, receive many industrial and domestic effluents from Namakkal and surrounding areas. Station 2 is located at the site of a small dam built for field irrigation and receives effluents from domestic sources, hospitals and other sources. The other stations are from normal flowing parts of the river (water runs faster in those areas), but they also receive additional inputs from sewage waters. In the same day of capture the animals were brought to the laboratory as soon as they were caught and dissected with clean instruments. The tissues of ten fish of the same species from the same station were pooled to make 3 subsamples. Gill, liver and kidney were put in Petri dishes to dry at 120°C until reaching a constant weight. Dry gill (0.423 mg), liver (0.307 mg) and muscle (0.682 mg) tissue was put into digestion flasks and 4 ml perholic acid and 8 ml nitric acid (Merk) were added. The digestion flasks were then put on a hot plate set to 130°C gradually increased until all materials were dissolved. Digests were diluted with distilled water appropriately in the range of the standards, which were prepared

from the stock standard solution of the metals (Merck). Metal concentrations in the samples were measured using a Perkin Elmer AS 3100 atomic absorption spectrophotometer and given as $\mu\text{g metal/g dry weight}$. Statistical Analysis of data was carried put with SPSS statistical package programs. Kristal-Wallis one way ANOVA was used to compare data among stations.

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SAMPLING AREA:

TABLE: 1. List of native places in Namakkal district and their availability of toxic heavy metals in a water samples.

S.NO	SAMPLING AREA	DATE &TIME	AVAILABILITY OF TOXIC METALS
1.	Kaliyanoor	28.02.2022	POSITIVE
2.	Pethampalayam	27.03.2022	NEGATIVE
3.	Kallankattu valasu	03.04.2022	NEGATIVE
4.	Puliyankadu	30.05.2022	POSITIVE
5.	Kullanaikkam palayam	30.03.2022	NEGATIVE
6.	Thatankuttai	26.06.2023	POSITIVE
7.	Uppukulam	30.07.2022	NEGATIVE
8.	Nallampalayam	03.08.2022	POSITIVE
9.	Elayampalayam	02.09.2022	NEGATIVE
10.	Avathipalayam	02.10.2022	NEGATIVE
11.	Sunnambu odai pond	22.02.2023	POSITIVE



Figure 2: collection of water samples in studying area

CHAPTER- III

RESULTS

The result concludes the contamination of poultry waste fresh water bodies in Namakkal district. The concentration and associated standard deviations of Lead, Copper, Chromium and Nickel in the Gills, Liver, Muscle and Kidney of *Labia rohita* from 5 stations in the Sunnambu odai pond in Namakkal are shown in Tables 1, 2, 3, 4 along with the result and statistical comparison was given. Figure 1 shows sampling stations in the Sunnambu odai pond. Cadmium concentrations varied significantly ($p < 0.05$) from station to station except in the muscle of *Labia rohita* (Table: 2). Station 2 generally showed the higher Lead concentrations varied significantly from station to station, except ($p > 0.05$) in the gill of *Labia rohita* (Table: 3). The higher lead concentrations were found in samples from Station 2, except in the muscles. Copper concentrations varied from station to station only in the gill liver of *Labia rohita* (Table: 4). None of the stations consistently showed the highest copper concentrations. Mean copper concentrations were higher in the gill of *Labia rohita*. Variations in chromium concentrations were significantly only in the gill and liver of *Labia rohita*. Station 2 showed the highest chromium concentrations in the tissues except in the muscle and gill of *Labia rohita*. Nickel concentrations varied significantly from station to station in the gill tissue of all fish species (Table: 5). Station 2 showed the highest nickel concentrations. Mean nickel concentrations were highest in the gill and liver tissue of *Labia rohita*.

TABLE: 2.TABULATION OF LEAD CONCENTRATION

STATION	GILL	LIVER	MUSCLE	KIDNEY
1	9.51	4.87	5.5	3.45
2	15.93	11.15	6.1	3.11
3	8.94	6.97	5.6	3.58
4	9.8	5.71	6.2	5.21
5	7.69	5.31	2.6	1.58
Mean	10.37	6.87	5.2	5.19

GRAPHICAL PRESENTATION OF LEAD CONCENTRATION

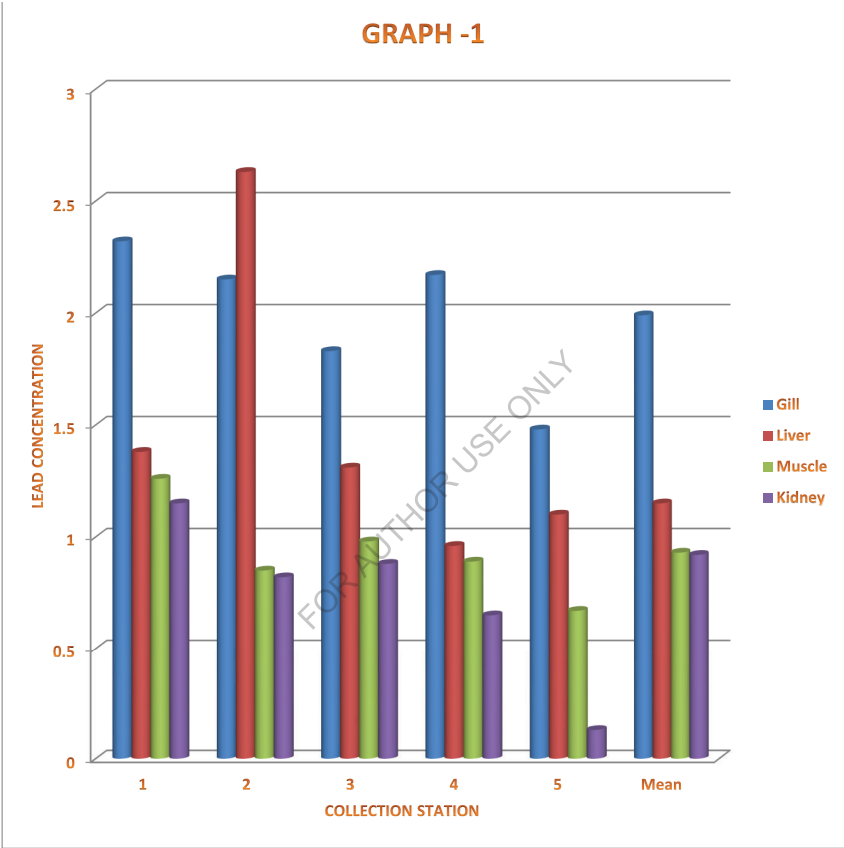


TABLE: 3.TABULATION OF COPPER CONCENTRATION

STATION	GILL	LIVER	MUSCLE
1	7.54	52.71	6.57
2	7.45	61.23	3.47
3	6.22	13.92	6.2
4	6.33	10.4	4.58
5	7.51	15.43	4.79
Mean	7.01	32.19	5.12

GRAPHICAL PRESENTATION OF COPPER CONCENTRATION

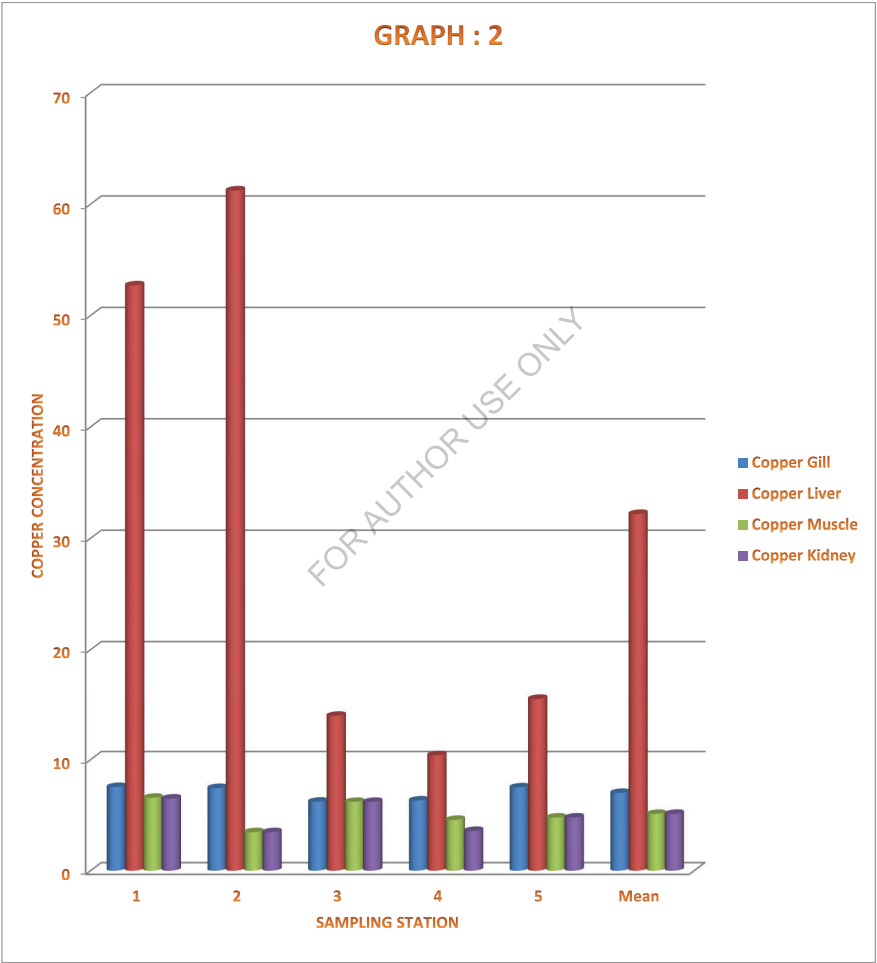


TABLE: 3.TABULATION OF CHROMIUM CONCENTRATION

STATION	GILL	LIVER	MUSCLE
1	7.54	52.71	6.57
2	7.45	61.23	3.47
3	6.22	13.92	6.2
4	6.33	10.4	4.58
5	7.51	15.43	4.79
Mean	7.01	32.19	5.12

GRAPHICAL PRESENTATION OF CHROMIUM CONCENTRATION

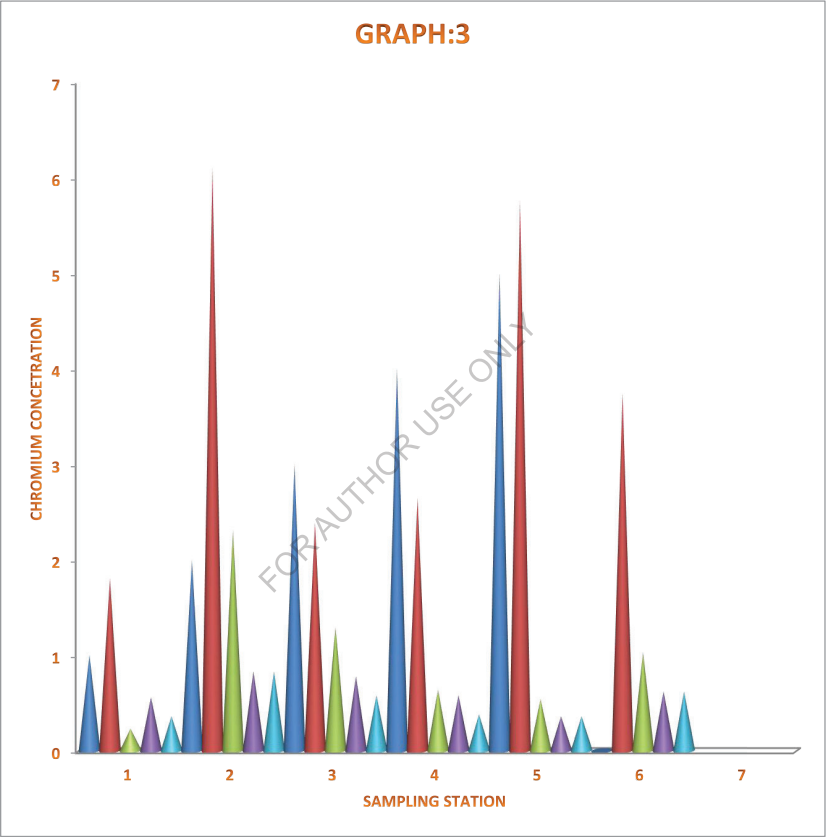
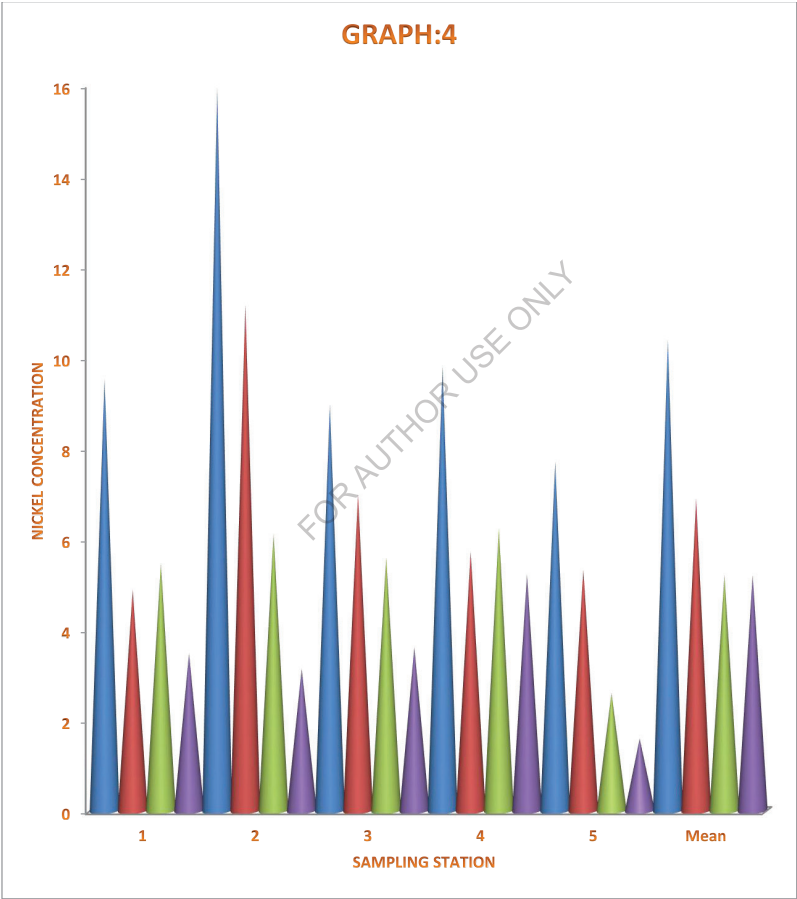


TABLE: 2.TABULATION OF NICKEL CONCENTRATION

STATION	GILL	LIVER	MUSCLE	KIDNEY
1	9.51	4.87	5.5	3.45
2	15.93	11.15	6.1	3.11
3	8.94	6.97	5.6	3.58
4	9.8	5.71	6.2	5.21
5	7.69	5.31	2.6	1.58
Mean	10.37	6.87	5.2	5.19

GRAPHICAL PRESENTATION OF NICKEL CONCENTRATION



Mean concentrations and associated standard deviations of lead (1), copper (2), chromium (3) and nickel (4) in the gill, liver and muscle tissue of *Labia rohita* caught at five stations in the kaliyanoor,Thattankuttai,Pethampalayam,Uppukulam,Kallankattuvalasu,Nallampalayam,Puliy ankadu,Avathipalayam,Kullanaikanpalayam.Sunambu odai, Sunnambu odai pond was final statistical differences (P Value) from station to station are indicated and the total mean concentration so metals in each tissue for each fish species are also given's=not significant (P>0.05), NA=not available.

FIGURE2:POULTRY FARMS NEAR NATIVE PLACESSUNNambu ODAI POND IN NAMAKKAL



CHAPTER IV

DISCUSSION

Initially we expected that heavy metal concentrations in the tissue of fresh water fish from Stations 4 and 5 would be highest as they are in the south of Sunnambu odai pond, which receives more untreated textile waste water from the river and surrounding environment. However, fish species from Station 2 displayed the highest metal concentrations in their tissues. An investigation was carried out to determine the source of high metal input to this station and the results showed that several textile industry and hospitals as well as domestic sources discharge untreated effluents in the vicinity of station 2. The results of this study indicate that fish can accumulate heavy metals efficiently in areas where direct inputs occur; the literature states that metal uptake from water is much higher than uptake from sediment (Goyer, 1991). The fact that fish from Stations 3-5 displayed lower metal concentrations than those from Stations 2 suggests that metals discharged in the vicinity of Station 2 are perhaps precipitated or adsorbed onto sediments due to interaction with some other compounds, so that fish at stations 3-5 are not exposed to the metals in water to the same degree as fish living at Station 2. From the results of this study, Station 1 (Sunnambu odai pond) appears to be the cleanest part of the river system, probably because it does not receive many pollutants from textile industrial and domestic sources. It is generally accepted that heavy metal uptake occurs mainly from water, food and sediment (bottom feeders and burrowing

animals the efficiency of metal uptake from contaminated water and food may differ in relation to ecological needs, metabolism, and the contamination gradients of water, food and sediment, as well as other factors such as salinity, temperature and interacting agents (Kalay et al., 1995). Dallinger and Kautzky (Tort and Torres, 1998) indicated that the uptake of cadmium, copper and zinc from food could be the main route of metal uptake by *Labia rohita*, although high levels of elimination through faeces occur. Kilgour (22) pointed out that the release of cadmium sediment can increase the cadmium concentration in the water and indicated that bottom-feeding and burrowing animals showed relatively high body concentrations of cadmium, although uptake from water was a more important route for other animals. Whatever the source of metals, the results of the present study showed that concentrations of cadmium, lead, copper, chromium and nickel were very high at nearly all stations when compared to data obtained from uncontaminated waters (Pagenkopf 1983). The levels of some metals were even higher than the acceptable values for human consumption designated by various health organizations (Lareeon *et al.*, 1985). Especially when one considers that stations chosen here are from areas which are important for commercial and amateur fishers and also for leisure (Stations 1 and 2), these results should serve as a warning. From the fish health point of view, the results of the present study may also be considered as an important warning signal. It is well known that heavy metals accumulated in substantially high levels can be very toxic for fish, especially

for young and eggs which are very sensitive to pollution (Glober et al., 1989). Target organs, such as the liver and gills, are metabolically active tissues and accumulate heavy metals in higher levels, as was observed in experimental and field studies. In the literature, heavy metal concentrations in the tissues of freshwater fish varies considerably among different studies, possibly due to differences in metal concentrations and chemical characteristics of water from which fish were sampled, ecological needs, metabolism and feeding patterns of fishes and also the season in which studies were carried out. Namakkal found the highest metal concentrations in the liver, followed by the gill and muscle in fishes from South Cauvery River indicated that no fish were caught at a heavily polluted station, possibly due to hindrance of aquatic life. The order of metal accumulation in the tissues of this study was also supported by the present results. Sunnambu odai Pond also found considerable differences in mercury concentrations in the muscle of different fish from the river in Namakkal, indicating that metabolic activities of different fish species may be an important factor in mercury accumulation. Dallinger and Kautzky also found similar distribution patterns of metals in *Labia rohita*. Highest concentrations of zn, pb, Cd, Cr and Ni were observed in the gill of *Labia rohita*, followed by the liver and muscle, while Cu concentrations was highest in the liver, followed by the gill and muscle. They also indicated that different concentrations of heavy metals in different fish species might be a result of different ecological needs, metabolism and feeding patterns. Lead concentrations

in the muscle of the *Labia rohita* and the from various aquatic areas of the Namakkal ranged up to 7.40 and 8.70 ppm, while cadmium levels ranged up to 1.88 and 1.07 ppm in the *Labia rohita* and Studied filed bioaccumulation of heavy metals in the tissues of *Labia rohita* exposed to contaminated water. They found that two different experimental stations of the pond showed different metal concentrations after 30 days, indicating stations contaminated heavily caused the fish to accumulate heavy metals in greater levels. They found metal accumulation in the tissues in the following order. Pb in the bone, copper in the spleen and kidney, Cr in the muscle and gills, nickel in the kidney, however, they did not study the liver, which is an important organ in metal accumulation, so these orders may be incomplete. Whole body cadmium and zinc concentrations, were found to be very high in fish from industrially contaminated area, highest cadmium and zinc concentrations being 13.60 and 820 ppm respectively). Sharif et al., studied metal concentrations in the edible parts of ten freshwater fish species from Bangladesh. The ranges of iron, nickel, copper and zinc concentrations were 31.8-296, 1.20-6.10, 23.31-48 and 33.01-286 ppm, respectively. The ranges of lead and cadmium levels were 0.29-10.05 and 0.04-0.13 ppm, respectively.

CHAPTER- V

SUMMARY & CONCLUSION

In this present study toxic effect of heavy metals on fresh water fishes it is clear that depending upon the textile pollutant water in the native places such as kaliyanoor, Thattankuttai, Pethampalayam, Uppukulam, Kallankattuvalasu, Nallampalayam, Puliyankadu, Avat hipalayam, Kullanaikanpalayam. Sunambu odai, Which is coming from varies Textile industries and hospitals, rural areas that directly affect the aquatic animals and human beings who consume the same? Certain heavy metal levels reached unacceptable levels for human consumption through fish consumption. Because high metal concentrations in tissue can have toxic effects on fish metabolism, it is important to consider the biological effects of contamination on fish health in the native places in Namakkal.

Usually all aquatic animals bear only certain amount of heavy metals when this goes beyond the limit this cause harmful effect directly to aquatic animals and indirectly to human beings.

finally create awareness of effect of heavy metals on fishes and toxic effect of consumer and fisherman our native place .

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